

Q1. Real-Time Detection System

The surveillance system uses three different object detection techniques: Viola-Jones, YOLO, and Deep Learning. The important factors for a surveillance system are accuracy and detection speed. Since the system has strict latency constraints, the method should detect objects quickly while still providing acceptable accuracy.

Method	Accuracy	Speed
Viola-Jones	78%	Fast
YOLO	92%	Moderate
Deep Learning	95%	Slow

Comparison

- ❖ **Viola-Jones** has an accuracy of 78% and operates at fast speed. It is suitable when the system has very limited computational resources and requires quick detection. However, its accuracy is comparatively low, which may result in missed or incorrectly detected objects.
- ❖ **YOLO** provides 92% accuracy with moderate speed. It gives a good balance between detection accuracy and processing speed. It is capable of detecting objects in real time and is commonly suitable for surveillance applications.
- ❖ **Deep Learning** provides the highest accuracy of 95%, but its speed is slow. It may require powerful hardware such as GPUs and can introduce higher latency. Therefore, it may not be suitable when immediate detection is necessary.

Evaluation and Selection

For a surveillance system operating under strict latency constraints, I would select YOLO. Although Deep Learning provides slightly higher accuracy, the difference between YOLO and Deep Learning is only 3 percentage points, while Deep Learning is significantly slower. Viola-Jones is faster, but its accuracy of 78% may not be acceptable for security-related applications. YOLO provides 92% accuracy with moderate processing speed, making it a practical compromise. In a surveillance system, detecting a suspicious person or object quickly is very important. A highly accurate model that responds too slowly may not be useful in real-time situations.

Deployment Conditions

The final choice also depends on the deployment environment:

- If the system has very low computational power, Viola-Jones may be considered.
- If the system requires real-time detection with good accuracy, YOLO is preferable.
- If latency is not important and powerful hardware is available, Deep Learning can be selected.

- For CCTV cameras, security monitoring, and real-time object detection, YOLO provides a good balance.

Conclusion

Therefore, YOLO is the most suitable method for this surveillance system because it provides high accuracy while maintaining moderate speed. It satisfies the practical requirement of real-time operation better than the other two methods.

Q2. Robust Object Detection

The performance of the three methods is given under three different environmental conditions: normal conditions, occlusion, and low-light conditions.

Method Normal Occlusion Low Light

Viola-Jones	80%	60%	55%
YOLO	90%	75%	70%
Deep Learning	95%	85%	80%

Comparison Under Normal Conditions

Under normal conditions, Deep Learning performs the best with 95% accuracy. YOLO achieves 90%, while Viola-Jones achieves 80%.

This shows that Deep Learning can extract more complex features from images and identify objects more accurately. Viola-Jones performs reasonably well but is less accurate than the other two methods.

Comparison Under Occlusion

Occlusion occurs when an object is partially hidden by another object.

- Viola-Jones: **60%**
- YOLO: **75%**
- Deep Learning: **85%**

Deep Learning performs best under occlusion. Its accuracy decreases by only 10 percentage points from normal conditions, whereas Viola-Jones decreases by 20 percentage points.

This indicates that Deep Learning has better ability to recognize partially visible objects.

Comparison Under Low-Light Conditions

Under low-light conditions:

- Viola-Jones: 55%
- YOLO: 70%
- Deep Learning: 80%

Again, Deep Learning provides the highest accuracy. Viola-Jones experiences a significant performance reduction in low-light environments.

This means Deep Learning is more robust when the lighting conditions are poor.

Overall Performance

The average accuracy of each method can be calculated as:

Viola-Jones:

YOLO:

Deep Learning:

Therefore, the overall ranking is:

1. **Deep Learning – 86.67%**
2. **YOLO – 78.33%**
3. **Viola-Jones – 65%**

Trade-Offs

Deep Learning provides the best robustness and accuracy, but it usually requires more computational resources and may have higher processing time.

YOLO provides a compromise between accuracy, robustness, and speed. It is more suitable when real-time performance is important.

Viola-Jones has the advantage of being fast and computationally less demanding, but its performance decreases significantly under occlusion and low-light conditions.

Conclusion

Deep Learning provides the best overall performance, with an average accuracy of approximately 86.67%. It is the best choice when accuracy and robustness are the main requirements.

However, if the application requires real-time processing and has limited hardware resources, YOLO may be a more practical choice. Thus, the final decision depends on whether accuracy or processing speed is more important.

Q3. Optical Flow Analysis

Optical flow is used to estimate the movement of objects between consecutive frames of a video. Two methods are available:

- **Method A:** Produces stable motion vectors but misses small motions.
- **Method B:** Detects fine motion but produces noisy vectors.

Comparison

❖ **Method A** produces stable and smooth motion vectors. This means that the detected movement is less affected by noise. However, its main disadvantage is that it may fail to detect small or subtle movements.

❖ **Method B** is more sensitive and can detect small movements. This is useful when even minor object movements are important. However, because it is highly sensitive, it also produces noisy motion vectors. These noisy vectors may result in incorrect movement estimation.

Factor	Method A	Method B
Motion accuracy	Good for large motion	Good for fine motion
Stability	High	Lower
Small motion detection	Poor	Excellent
Noise sensitivity	Low	High
Suitable for	Stable tracking	Detailed motion detection

Accuracy

- Method B can provide better accuracy when the application requires detection of small and detailed movements. For example, it can be useful for detecting small hand movements or subtle object movements.
- Method A may be more accurate for applications involving larger and continuous movements because its vectors are stable.

Stability

Method A is better in terms of stability. Its motion vectors are less affected by random variations, so tracking can be smoother.

Method B may produce sudden or irregular vectors because of noise, which can make tracking unstable.

Noise Sensitivity

Method A is less sensitive to noise. This makes it suitable for environments where the video contains disturbances.

Method B is more sensitive to noise. Therefore, additional filtering or smoothing may be required before using its output.

Application Requirements

The choice depends on the application.

For example:

- For vehicle tracking, stable motion estimation is important, so Method A may be preferred.
- For gesture recognition, small movements are important, so Method B may be preferred.
- For surveillance, Method A may be useful when stable tracking is more important than detecting tiny movements.
- For detailed motion analysis, Method B can provide more useful information.

Conclusion

For general motion tracking in dynamic scenes, I would prefer Method A if stable and reliable tracking is the main requirement. It produces less noisy vectors and provides consistent motion estimation.

However, if the application specifically requires detecting small and fine movements, Method B is more suitable. Its noise problem can be reduced using filtering and smoothing techniques. Therefore, the best method depends on the application: Method A for stable tracking and Method B for fine-motion detection.

Q4. Model Generalization and Overfitting

The performance of two models is given below:

Model Training Accuracy Testing Accuracy

A	98%	70%
B	90%	88%

The difference between training and testing accuracy is important for understanding **generalization and overfitting**.

Model A

Model A has:

- Training accuracy = **98%**
- Testing accuracy = **70%**

The model performs extremely well on the training data but its performance decreases significantly on unseen testing data.

The difference is:

This is a very large gap.

This indicates that Model A has likely overfitted the training data. It may have learned specific patterns or noise from the training dataset instead of learning general patterns that can be applied to new data.

Although 98% training accuracy looks very good, the testing accuracy of only 70% indicates poor real-world generalization.

Model B

Model B has:

- Training accuracy = **90%**
- Testing accuracy = **88%**

The difference is:

This is a very small gap.

The model performs almost equally well on training and testing data. Therefore, Model B has a much better ability to generalize to unseen data.

It has a lower overfitting risk compared with Model A.

Comparison

Factor	Model A	Model B
Training Accuracy	98%	90%
Testing Accuracy	70%	88%
Accuracy Gap	28%	2%
Overfitting Risk	High	Low
Generalization	Poor	Good
Real-World Reliability	Lower	Higher

Generalization Ability

Generalization means the ability of a machine learning model to perform well on new and unseen data.

Model B has better generalization because its training and testing performances are very close. Model A has poor generalization because its performance falls from 98% during training to only 70% during testing.

Overfitting Risk

- Model A has a high overfitting risk. It may have memorized the training data rather than learning useful general patterns.
- Model B has a low overfitting risk because the difference between training and testing accuracy is only 2%.
- Selection for Real-World Deployment
- I would select Model B for real-world deployment.

- Even though Model A has a higher training accuracy, the testing accuracy is much lower. In a real-world application, the model will encounter data that it has never seen before. Therefore, testing accuracy and generalization are more important than simply achieving very high training accuracy.
- Model B achieves 88% testing accuracy, which is significantly higher than Model A's 70%.

Conclusion

Model B is more reliable for real-world deployment. It has better generalization, a smaller training-testing gap, and a lower risk of overfitting. Model A should be improved using techniques such as regularization, data augmentation, cross-validation, or increasing the training dataset before deployment.

Q5. System Performance under Constraints

The three models have different accuracy, latency, and cost.

Model Accuracy Latency Cost

A	88%	50 ms	Low
B	92%	80 ms	Moderate
C	95%	120 ms	High

For a real-time application, all three factors are important. A model should not only have high accuracy but should also respond quickly and have a reasonable deployment cost.

Model A

Model A provides **88% accuracy**, **50 ms latency**, and **low cost**.

Its biggest advantage is its low latency. A response time of 50 ms is suitable for many real-time applications. It is also economical to deploy.

However, its accuracy is the lowest among the three models.

Model B

Model B provides **92% accuracy**, **80 ms latency**, and **moderate cost**.

It improves accuracy by 4 percentage points compared with Model A. Its latency is higher at 80 ms, but it can still be suitable for many real-time applications.

Its moderate cost is also reasonable for practical deployment.

Model C

Model C has the highest accuracy of **95%**, but its latency is **120 ms** and its cost is **high**.

Although it provides the best accuracy, the high latency can become a problem for applications that require immediate responses.

The high cost can also make large-scale deployment more difficult.

Comparison

Factor	Model A	Model B	Model C
Accuracy	88%	92%	95%
Latency	50 ms	80 ms	120 ms
Cost	Low	Moderate	High
Real-Time Suitability	Very Good	Good	Lower

Factor	Model A	Model B	Model C
Overall Balance	Good	Best	Expensive

Evaluation

If only accuracy is considered, Model C is the best because it achieves 95%.

If only latency and cost are considered, Model A is the best because it has the lowest latency and cost.

However, a practical real-time application needs a balance between all three factors.

Model B provides 92% accuracy, which is considerably better than Model A, while its latency of 80 ms is still reasonable for many real-time applications. Its moderate cost is also more manageable than the high cost of Model C.

Therefore, Model B provides the best overall balance.

Practical Deployment Conditions

For example, consider a real-time surveillance or monitoring application. The system must provide accurate results without taking too long to respond.

- Model A may be selected when the budget is very limited.
- Model B is suitable when accuracy, speed, and cost all need to be balanced.
- Model C may be selected for applications where maximum accuracy is more important than latency and cost.

Conclusion

I would select Model B for a general real-time application.

It provides 92% accuracy, 80 ms latency, and moderate cost. Although Model C is more accurate, its 120 ms latency and high cost may not be practical. Model A is faster and cheaper, but its accuracy is lower.

Therefore, Model B gives the best balance between accuracy, latency, and cost, making it the most practical choice for real-world deployment.